Question 1

```
Tilix: frank@cis106: ~
1: frank@cis106: ~
bastet/stable 0.43.2-2 amd64
  ncurses Tetris clone with a bastard algorithm

ltris/stable 2.0.3-1 amd64
  very polished Tetris clone with CPU opponents

tint/stable 0.07 amd64
  Tetris clone for text based terminal

-- MOST: *stdin*                                                    (1,1)100%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```

```
3: frank@cis106: ~
bastet/stable 0.43.2-2 amd64
  ncurses Tetris clone with a bastard algorithm

blockattack/stable 2.9.0-1+b1 amd64
  puzzle game inspired by Tetris

blockout2/stable 2.5+dfsg1-2 amd64
  Tetris like game (3D-tetris)

blocks-of-the-undead/stable 1.0-7 amd64
  Tetris Attack clone with spooky undertones

blocks-of-the-undead-data/stable 1.0-7 all
  Tetris Attack clone with spooky undertones - data files

bsdgames/stable 2.17-35 amd64
-- MOST: *stdin*                                                    (1,1)0%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```

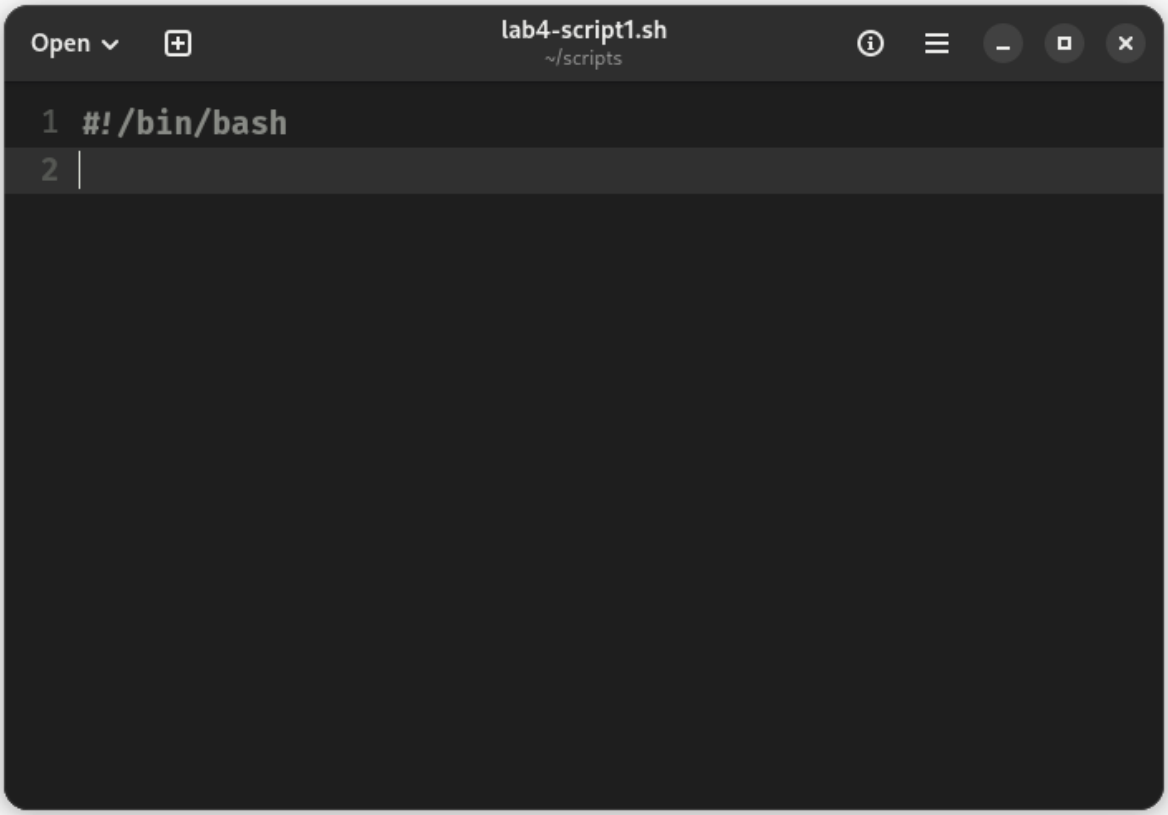
```
2: frank@cis106: ~
games-tetris/stable 5 all
  Debian's tetris-like games

vitetris/stable 0.59.1-3 amd64
  Virtual terminal *tris clone

-- MOST: *stdin*                                                    (1,1)100%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```

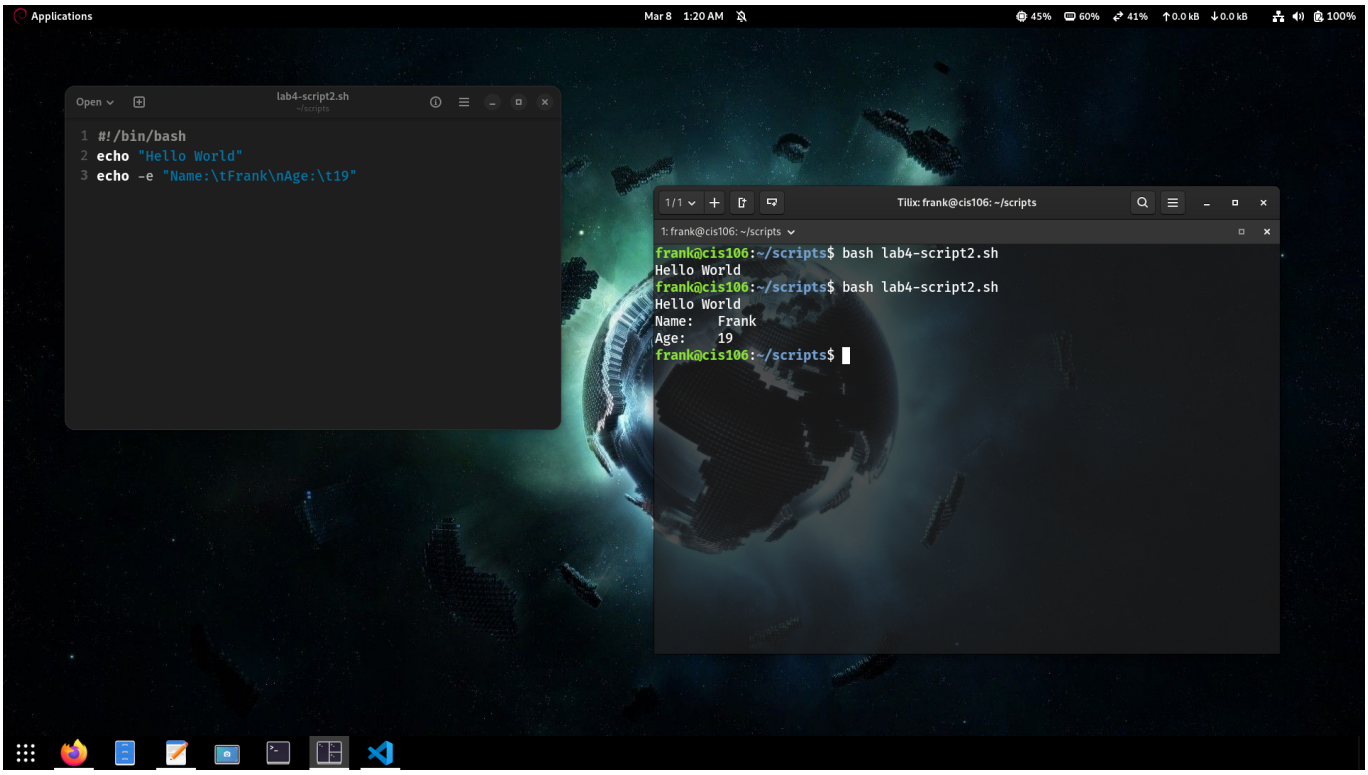
```
Tilix: frank@cis106: ~
1: frank@cis106: ~
frank@cis106:~$ sudo apt install blockout2+ cuyo+ blockattack-
The following packages were automatically installed and are no longer required:
  libfmt10  libopusfile0  libphysfs1  libsdl2-image-2.0-0  libsdl2-mixer-2.0-0  libsdl2-ttf-2.0-0  libxmp4  linux-image-6.12.63+deb13-amd64
Use 'sudo apt autoremove' to remove them.

Installing:
  blockout2  cuyo

Installing dependencies:
  cuyo-data  libmikmod3  libsdl-mixer1.2  libsdl1.2debian

REMOVING:
  blockattack

Summary:
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 6
  Download size: 8,078 kB
  Space needed: 4,102 kB / 27.0 GB available

Continue? [Y/n] Y
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libsdl1.2debian amd64 1.2.68-3 [85.9 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libsdl-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]
Fetched 8,078 kB in 1s (13.5 MB/s)
(Reading database ... 201154 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:amd64.
(Reading database ... 201127 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_amd64.deb ...
Unpacking libmikmod3:amd64 (3.3.13-1) ...
Selecting previously unselected package libsdl1.2debian:amd64.
Preparing to unpack .../1-libsdl1.2debian_1.2.68-3_amd64.deb ...
Unpacking libsdl1.2debian:amd64 (1.2.68-3) ...
Selecting previously unselected package libsdl-mixer1.2:amd64.
Preparing to unpack .../2-libsdl-mixer1.2_1.2.12-19+b1_amd64.deb ...
```

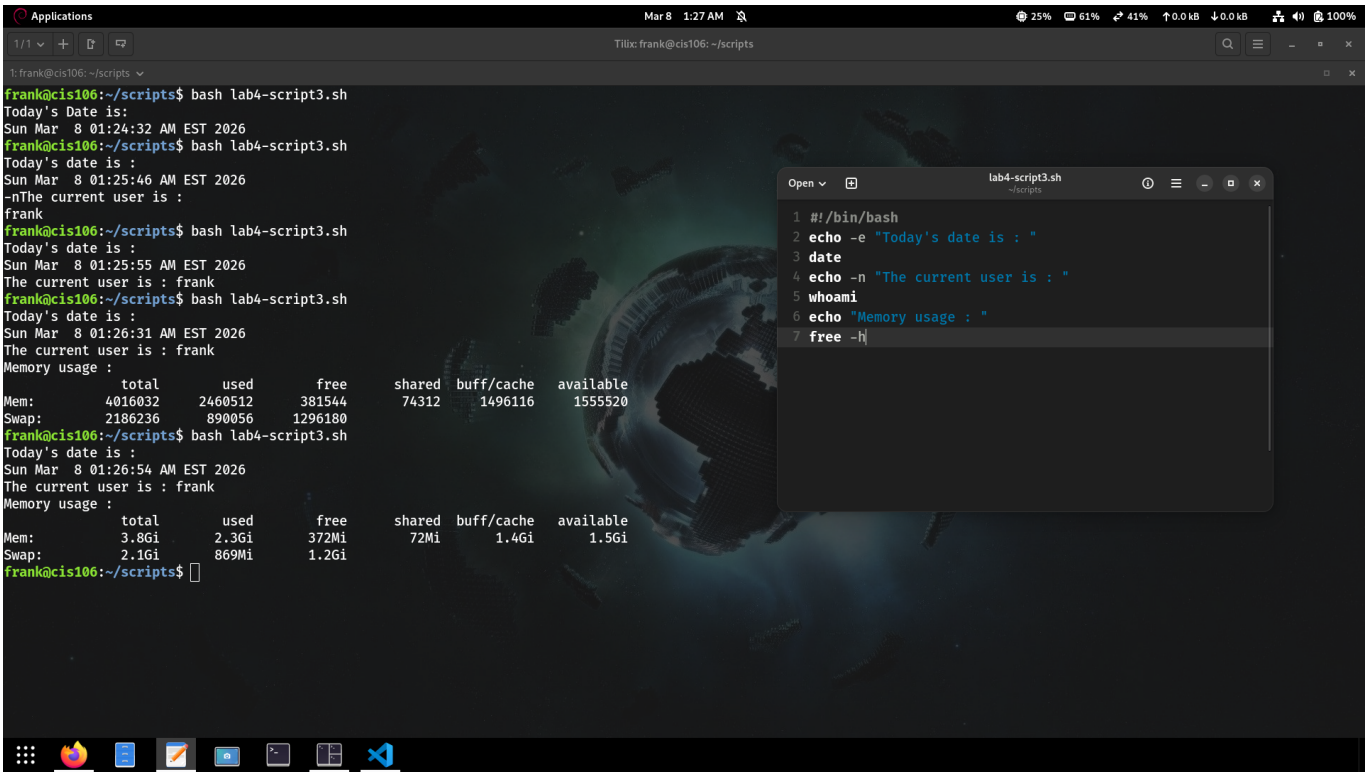
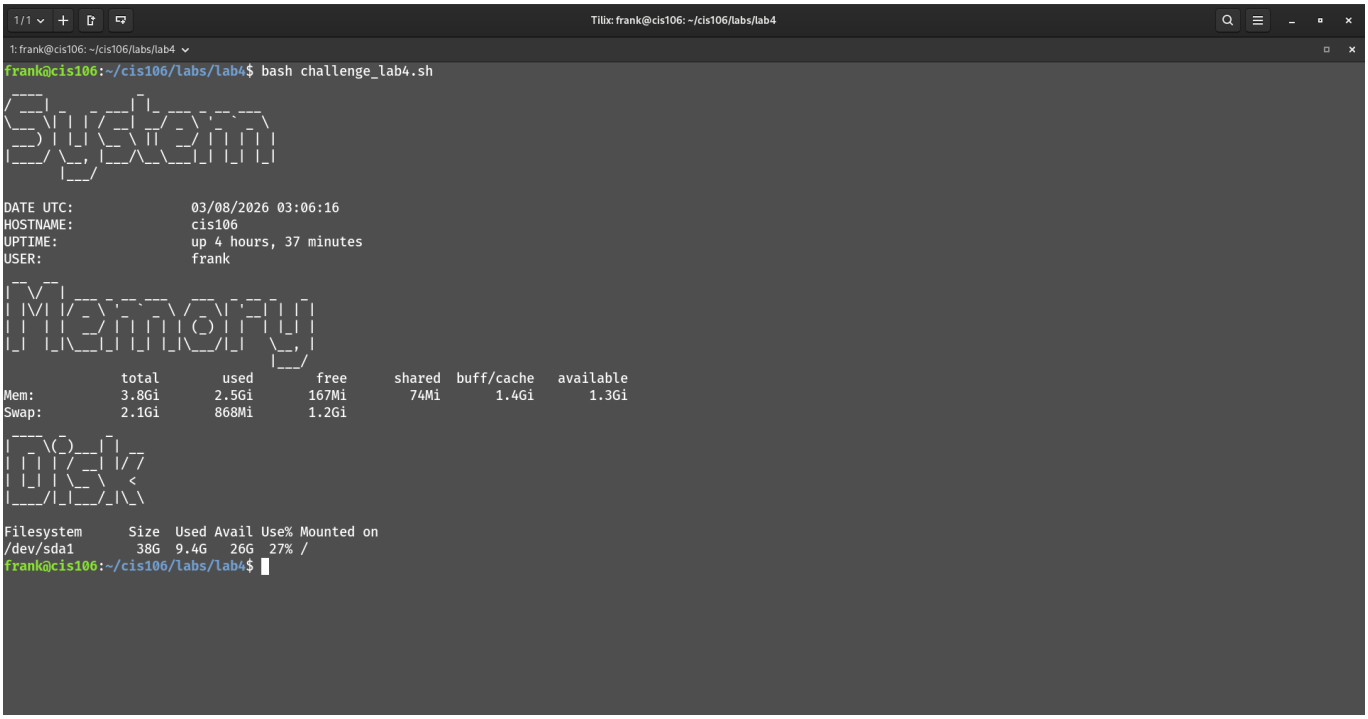
```
Tilix: frank@cis106: ~
1: frank@cis106: ~
frank@cis106:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libfmt10  libopusfile0  libphysfs1  libsdl2-image-2.0-0  libsdl2-mixer-2.0-0  libsdl2-ttf-2.0-0  libxmp4  linux-image-6.12.63+deb13-amd64
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 6
  Freed space: 489 kB

Continue? [Y/n] Y
(Reading database ... 202224 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
frank@cis106:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libfmt10  libopusfile0  libphysfs1  libsdl2-image-2.0-0  libsdl2-mixer-2.0-0  libsdl2-ttf-2.0-0  libxmp4  linux-image-6.12.63+deb13-amd64
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 6
  Space needed: 0 B / 27.0 GB available

Continue? [Y/n] Y
(Reading database ... 202216 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1) ...
frank@cis106:~$
```

```
Tilix: frank@cis106: ~
1: frank@cis106: ~
frank@cis106:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
  cuyo-data  libfmt10  libopusfile0  libphysfs1  libsdl2-image-2.0-0  libsdl2-mixer-2.0-0  libsdl2-ttf-2.0-0  libxmp4  linux-image-6.12.63+deb13-amd64

Summary:
  Upgrading: 0, Installing: 0, Removing: 9, Not Upgrading: 6
  Freed space: 118 MB

Continue? [Y/n] Y
(Reading database ... 202216 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12-4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:amd64 (2.8.8+dfsg-1) ...
Removing libsdl2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Removing linux-image-6.12.63+deb13-amd64 (6.12.63-1) ...
/etc/kernel/postrm.d/initramfs-tools:
update-initramfs: Deleting /boot/initrd.img-6.12.63+deb13-amd64
/etc/kernel/postrm.d/zz-update-grub:
Generating grub configuration file ...
Found background image: /usr/share/images/desktop-base/desktop-grub.png
Found linux image: /boot/vmlinuz-6.12.73+deb13-amd64
Found initrd image: /boot/initrd.img-6.12.73+deb13-amd64
Found linux image: /boot/vmlinuz-6.12.69+deb13-amd64
Found initrd image: /boot/initrd.img-6.12.69+deb13-amd64
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 6
frank@cis106:~$
```

Question2


Open
lab4-script1.sh
~/scripts
#!/bin/bash


Applications
Mar 8 1:20 AM
45% 60% 41% 0.0 kB 0.0 kB 100%
Open
lab4-script2.sh
~/scripts
#!/bin/bash
echo "Hello World"
echo -e "Name:\tFrank\nAge:\t19"
1/1
Tilix: frank@cis106: ~/scripts
1: frank@cis106: ~/scripts
frank@cis106:~/scripts$ bash lab4-script2.sh
Hello World
frank@cis106:~/scripts$ bash lab4-script2.sh
Hello World
Name: Frank
Age: 19
frank@cis106:~/scripts$



Challenge Question